

JAVA PROGRAMMING

PROJECT – 3

ATM

CODE:

ATM.JAVA

```
import java.util.Scanner;

public class ATM {
    private BankSystem bankSystem;    // Fixed class name capitalization
    private Scanner sc;

    public ATM() {
        bankSystem = new BankSystem();
        sc = new Scanner(System.in);
    }

    public void start() {
        int choice;
        do {
            System.out.println("\n===== ATM MANAGEMENT =====");
            System.out.println("1. Create Account");
            System.out.println("2. Login");
            System.out.println("3. Show All Accounts");
            System.out.println("4. Exit");
            System.out.print("Enter Choice : ");
            choice = sc.nextInt();

            switch (choice) {
                case 1: createAccount(); break;
                case 2: loginMenu(); break;
                case 3: bankSystem.showAllAccounts(); break;
                case 4: System.out.println("\nThank You for Using ATM!"); break;
                default: System.out.println("Invalid Choice!");
            }
        } while (choice != 4);
    }

    private void createAccount() {
        System.out.println("\n===== CREATE ACCOUNT =====");
        sc.nextLine(); // consume newline
        System.out.print("Enter Name : ");
        String name = sc.nextLine();
    }
}
```

```
        System.out.print("Enter Account Number : ");
        int accountNumber = sc.nextInt();
        System.out.print("Create PIN : ");
        int pin = sc.nextInt();
        System.out.print("Initial Deposit : ");
        double balance = sc.nextDouble();

        bankSystem.createAccount(name, accountNumber, pin, balance);
    }

    private void loginMenu() {
        System.out.println("\n===== LOGIN =====");
        System.out.print("Account Number : ");
        int accountNumber = sc.nextInt();
        System.out.print("PIN : ");
        int pin = sc.nextInt();

        BankAccount account = bankSystem.login(accountNumber, pin);
        if (account != null) {
            userMenu(account);
        }
    }

    private void userMenu(BankAccount account) {
        int choice;
        do {
            System.out.println("\nWelcome " + account.getName());
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Check Balance");
            System.out.println("4. Transaction History");
            System.out.println("5. Change PIN");
            System.out.println("6. Logout");
            System.out.print("Enter Choice : ");
        } while (choice < 6);
    }
}
```

```

        choice = sc.nextInt();
        switch (choice) {
            case 1:
                System.out.print("Enter Amount : ");
                double deposit = sc.nextDouble();
                account.deposit(deposit);
                break;
            case 2:
                System.out.print("Enter Amount : ");
                double withdraw = sc.nextDouble();
                account.withdraw(withdraw);
                break;
            case 3:
                System.out.println("Current Balance : ₹" + account.getBalance());
                break;
            case 4:
                account.showTransactionHistory();
                break;
            case 5:
                System.out.print("Enter old PIN : ");
                int oldPin = sc.nextInt();
                System.out.print("Enter New PIN : ");
                int newPin = sc.nextInt();
                account.changePin(oldPin, newPin);
                break;
            case 6:
                System.out.println("Logged Out Successfully.");
                break;
            default:
                System.out.println("Invalid Choice!");
        }
    } while (choice != 6);
}

```

BANKACCOUNT.JAVA

```
import java.util.ArrayList;

public class BankAccount {
    private String name;
    private int accountNumber;
    private int pin;
    private double balance;
    private ArrayList<String> transactionHistory;

    public BankAccount(String name, int accountNumber, int pin, double balance) {
        this.name = name;
        this.accountNumber = accountNumber;
        this.pin = pin;
        this.balance = balance;
        transactionHistory = new ArrayList<>();
        transactionHistory.add("Account Created with Balance : ₹" + balance);
    }

    public String getName() {
        return name;
    }

    public int getAccountNumber() {
        return accountNumber;
    }

    public boolean validatePin(int enteredPin) {
        return pin == enteredPin;
    }

    public double getBalance() {
        return balance;
    }

    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
        }
    }
}
```

```

        transactionHistory.add("Deposited : ₹" + amount);
        System.out.println("Deposit Successful.");
    } else {
        System.out.println("Invalid Amount.");
    }
}

public void withdraw(double amount) {
    if (amount <= 0) {
        System.out.println("Invalid Amount.");
        return;
    }
    if (amount > balance) {
        System.out.println("Insufficient Balance.");
        return;
    }
    balance -= amount;
    transactionHistory.add("Withdrawn : ₹" + amount);
    System.out.println("Withdrawal Successful.");
}

public void changePin(int oldPin, int newPin) {
    if (pin == oldPin) {
        pin = newPin;
        transactionHistory.add("PIN Changed");
        System.out.println("PIN Changed Successfully.");
    } else {
        System.out.println("Wrong Old PIN.");
    }
}
}

```

```

    public void showTransactionHistory() {
        System.out.println("\n===== Transaction History =====");
        for (String transaction : transactionHistory) {
            System.out.println(transaction);
        }
    }
}

```

BANKSYSTEM.JAVA

```
import java.util.ArrayList;

public class BankSystem {
    private ArrayList<BankAccount> accounts;

    public BankSystem() {
        accounts = new ArrayList<>();
    }

    public void createAccount(String name, int accountNumber, int pin, double balance) {
        if (findAccount(accountNumber) != null) {
            System.out.println("Account Number already exists!");
            return;
        }
        BankAccount account = new BankAccount(name, accountNumber, pin, balance);
        accounts.add(account);
        System.out.println("=====");
        System.out.println("Account Created Successfully!");
        System.out.println("Welcome " + name);
        System.out.println("=====");
    }

    public BankAccount findAccount(int accountNumber) {
        for (BankAccount account : accounts) {
            if (account.getAccountNumber() == accountNumber) {
                return account;
            }
        }
        return null;
    }

    public BankAccount login(int accountNumber, int pin) {
        BankAccount account = findAccount(accountNumber);
        if (account == null) {
            System.out.println("Account Not Found!");
            return null;
        }
    }
}
```

```

3         if (account.validatePin(pin)) {
4             System.out.println("\nLogin Successful!");
5             System.out.println("Welcome " + account.getName());
6             return account;
7         } else {
8             System.out.println("Invalid PIN!");
9             return null;
10        }
11    }
12
13    public void showAllAccounts() {
14        if (accounts.isEmpty()) {
15            System.out.println("No Accounts Available.");
16            return;
17        }
18        System.out.println("\n===== Account List =====");
19        for (BankAccount account : accounts) {
20            System.out.println("Name : " + account.getName());
21            System.out.println("Account Number : " + account.getAccountNumber());
22            System.out.println("Balance : ₹" + account.getBalance());
23            System.out.println("-----");
24        }
25    }
26 }

```

MAIN.JAVA

```

1 public class Main {
2     public static void main(String[] args) {
3         ATM atm = new ATM();
4         atm.start();
5     }
6 }

```

TRANSACTION.JAVA

```

public class Transaction {
    private String type;
    private double amount;

    public Transaction(String type, double amount) {
        this.type = type;
        this.amount = amount;
    }

    public String getType() {
        return type;
    }

    public double getAmount() {
        return amount;
    }

    @Override
    public String toString() {
        return type + " : ₹" + amount;
    }
}

```

OUTPUT:

```

PS C:\Users\Akshitha Reddy\Desktop\New folder (3)\DecodeLabs_Java_P3> javac Main.java
PS C:\Users\Akshitha Reddy\Desktop\New folder (3)\DecodeLabs_Java_P3> java Main.java

===== ATM MANAGEMENT =====
1. Create Account
2. Login
3. Show All Accounts
4. Exit
Enter Choice : 1

===== CREATE ACCOUNT =====
Enter Name : akshitha
Enter Account Number : 7890
Create PIN : 1234
Initial Deposit : 5678
=====
Account Created Successfully!
Welcome akshitha
=====

```